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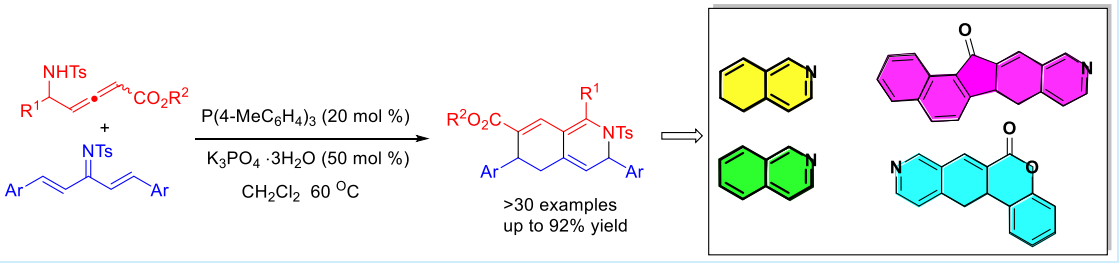
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Phosphine-Catalyzed Sequential [3+3]/Aza-6π-Electrocyclization Reaction of Cross-Conjugated Azatrienes and δ-Sulfonamido-Allenoates

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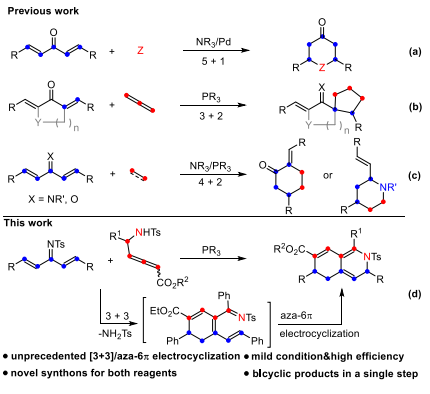
ABSTRACT: An unprecedented sequential [3+3]/aza-6π-electrocyclization between cross-conjugated azatrienes and δ-sulfonamido-allenoates, catalyzed by phosphine, has been developed, which provides efficient and facile access to highly functionalized tetrahydroisoquinoline derivatives. The products can be easily transformed into (dihydro)isoquinolines and their fused polycyclic compounds. The reactivity of both azatrienes and δ-sulfonamido-allenoates in this text, acting as a five-atom unit, is unique in the phosphine-catalyzed annulations of allenoates.

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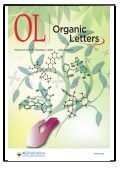
For decades, methodologies developed by phosphine-catalyzed reactions, especially those between allenoates and various electrophiles, have become a powerful and facile tool for the construction of complex structures.[1] However, the explosive development of phosphine catalysis has been greatly impeded in the past two years most likely due to the "old-patterned" reaction modes and synthons.[2] Our group has been dedicated to the exploration of phosphine catalysis, aiming at not only valuable skeletons formed in new reactions[3] but also the discovery of brand-new allenoates[4] that may act as totally different building blocks and lead to novel reaction modes, which, in our opinion, may reveal a promising breakthrough for the development of phosphine catalysis in the future.

Inspired by the unique "trinucleophilic site" mode of the δ-sulfonamido-allenoate developed in our previous work,[4a] we envisioned that more interesting chemistry could be developed with this novel allenoate. Cross-conjugated heterotrienes, as the heterosubstituted derivatives of dendralenes, captured our attention. Due to their unique ability to undergo sequential addition reactions, dendralenes are versatile building blocks for the quick generation of complex structures.[5] For instance, a cross-conjugated oxatriene was found to be a readily available Michael acceptor that is prone to undergo two additions. In 2011, Yan and co-workers disclosed an efficient synthesis of optically active cyclohexanones through asymmetric organo-catalytic double-conjugate addition of malononitrile.[6] Similarly, when a wide variety of nucleophilic reagents were adopted, cascade additions occurred and provided diverse cyclic compounds through [5+1] annulations (Scheme 1, a).[7,8] In addition, dibenzylidene acetone was also able to participate in the [3+2][9] and [4+2][10] reactions with allenoates or electron deficient olefins under organocatalysis (Scheme 1, b and c, X =

Scheme 1. Reactions of Cross-Conjugated Heterotrienes



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O). On the contrary, there have been few reports about cross-conjugated azatrienes. In general, cross-conjugated azatrienes bearing an electron-withdrawing sulfonyl or benzoyl group on the nitrogen atom underwent, on heating or in the presence of amine catalysts, an inverse electron demand hetero-Diels–Alder reaction with electron-rich dienophiles or aldehydes (Scheme 1, c, X = NR′).[11,12] Chi and co-workers reported a DMAP-catalyzed reaction of α-chloro acetic ester that provided attractive access to pyridines.[12b] Apart from that, Leung developed a palladacycle-catalyzed addition of Ph_2PH to α,β-unsaturated imines while there was one case of azatrienes involved.[13] Among all of these cases, only single-ring compounds were formed on the basis of a desymmetrization strategy via simple addition reactions, while the potential of cross-conjugated azatrienes to undergo sequential annulation to construct valuable fused-ring products remains challenging.

By employing novel δ-sulfonamido-allenoates, we hypothesized that cross-conjugated heterotrienes would undergo a rare [3+3] annulation, and the intermediate formed will be involved in an intramolecular aza-6π-electrocyclization, affording a bicyclic skeleton in one step. To the best of our knowledge, there has no report about aza-6π-electrocyclization involved in the annulation between allenoates and electrophiles catalyzed by phosphine. Herein, we developed the first sequential [3+3]/aza-6π-electrocyclization reaction that offers a general method for the efficient construction of polysubstituted hydroisoquinolines (Scheme 1, d). Moreover, the products could be easily transformed into aromatic bicyclic rings. Hydroisoquinoline and isoquinoline derivatives are frequently found in pharmaceuticals and bioactive natural products, as well as organocatalysts (Figure 1).[14−18] For

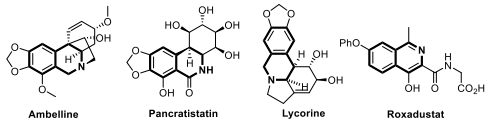


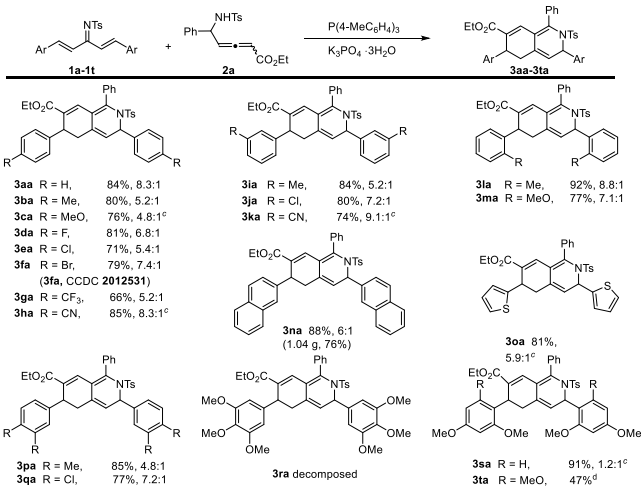
Figure 1. Representative bioactive molecules containing a (hydro)-isoquinoline skeleton.

example, Ambelline[15] and pancratistatin[16] are known to exhibit pronounced cytostatic activity. Roxadustat[18] (also known as FG-4592) is an orally bioavailable hypoxia-inducible factor (HIF) prolyl hydroxylase inhibitor that promotes coordinated erythropoiesis.

Initially, we started the approach by employing δ-sulfonamido-allenoate **2b** and cross-conjugated azatriene **1a** or dibenzylidene acetone (dba) **4** in the presence of PCy_3 and K_2CO_3 (for optimization of the reaction conditions, see Table S1). Gratifyingly, the reaction between **1a** and **2b** in chloroform delivered the desired product **3ab** in 24% yield with a diastereomeric ratio (dr) of 10: l. After conditions had been screened, the use of $P(4\text{-}MeC_6H_4)_3$ (20 mol %) with $K_3PO_4{\cdot}3H_2O$ (50 mol %) as a base in 1 mL of dichloromethane at 60 °C (sealed tube immersed in an oil bath) proved to be optimal.

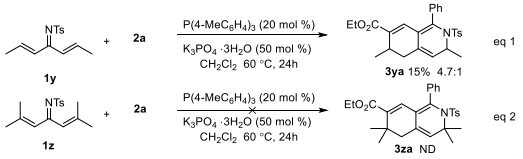
With the optimized condition in hand, we then turned to explore the generality of this novel strategy for the construction of highly substituted hydroisoquinolines. At first, we set out to investigate the scope of cross-conjugated azatrienes. Substrates **1a**–**1h** bearing an electron-donating group (Me and OMe) or an electron-withdrawing group (CF_3 and CN) as well as a halide (F, Cl, and Br) at the *para* position of both phenyl rings led to **3aa**–**3ha** in good yields with dr ranging from 4.8:1 to 8.3:1 (Scheme 2). The structure and relative configuration of

Scheme 2. Scope of Cross-Conjugated Azatrienes[a,b]



[a]Optimal conditions: **1** (0.1 mmol), **2a** (0.12 mmol), $P(4\text{-}MeC_6H_4)_3$ (20 mol %), $K_3PO_4{\cdot}3H_2O$ (50 mol %), in CH_2Cl_2 (1.0 mL) at 60 °C (sealed tube) for 24 h. [b]Combined yields of diastereomeric isomers are given. dr refers to the ratio of the major isomer (*anti*) to the minor isomer (*syn*). [c]dr determined through ^1H NMR. [d]Only the major isomer was obtained.

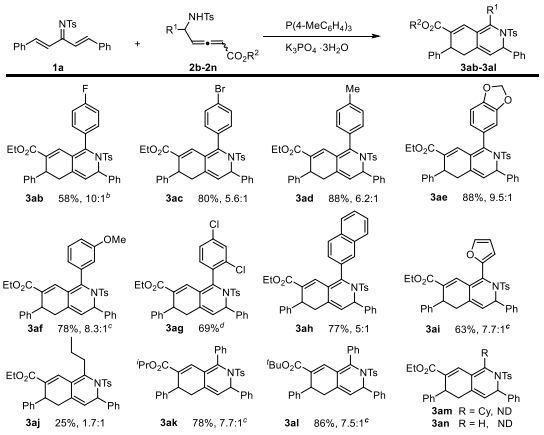
the major isomer (*anti*) were confirmed by X-ray crystallographic analysis (**3fa**, CCDC 2012531).[20] Similarly, the reaction between δ-sulfonamido-allenoate **2a** and cross conjugated aza-trienes **1i**–**1k** with different substituents at the *meta* position of the phenyl ring went smoothly to afford the desired hydroisoquinolines with acceptable diastereoselectivity. There is no obvious electric effect observed in these results. Surprisingly, products **3la** and **3ma** were well tolerated for the reaction, even in higher yields (91% and 77%, respectively). Moreover, it was found that azatrienes **1** containing a 2-naphthyl or heteroaryl group (thio) could also be used in this reaction with similar results. Product **3na** could be prepared on a gram scale under the optimal condition, and nearly 50% catalyst could be recovered. It is noteworthy that substrates **1** with multiple substitutions on the phenyl group in this methodology were provided with comparable yields (**3pa**–**3ta**). Specifically, product **3ra** with a 3,4,5-tri-MeO substitution on both phenyl rings was not stable during purification on silica gel. Aryl groups with larger steric groups were compatible with this sequential annulation; even 2,4,6-tri-MeO-substituted substrate **1t** afforded the polyphenyl tetrahydroisoquinoline derivative. When we changed the aryl group into an aliphatic substitution (methyl for **1y**), the corresponding product was also obtained (**3ya**, 15%, eq 1), which revealed the feasibility of aliphatic substituted hydroisoquinolines, while **1z** exhibited no reactivity because of bulky hindrance (67% recovered, eq 2).



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To further expand the scope of this methodology, we then investigated the synthesis of hydroisoquinolines with respect to different δ-sulfonamido-allenoates **2**. As shown in Scheme 3,

Scheme 3. Scope of δ-Sulfonamido-Allenoates 2[a,b]



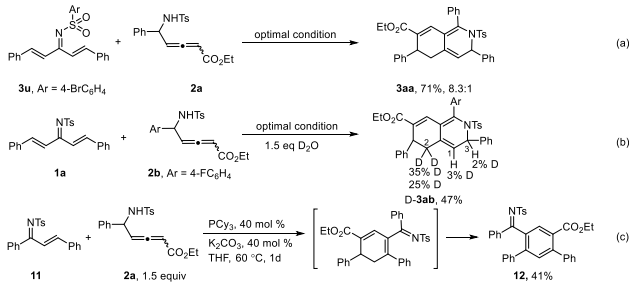
[a]Reaction carried out under optimal conditions. [b]Combined yields of diastereomeric isomers are given. dr refers to the ratio of the major isomer (*anti*) to the minor isomer (*syn*). [c]dr determined through ^{1}H NMR.

the current reaction showed good tolerance to the aryl group with *para* and *meta* substituents, giving **3ab**−**3af** in moderate to good yields. It should be noted that allenoate **2g**, which had a dichlorophenyl group, did participate in the reaction. However, the corresponding product **3ag** was obtained as a mixture with at least three pairs of isomers because of the extra axial chirality along with the 2,4-dichlorophenyl group. Moreover, allenoates **2h** and **2i** derived from 2-naphth- and furylaldehyde, respectively, were injected into the reaction mixture under the optimal condition, leading to the construction of hydroisoquinoline derivatives in moderate yields (77% and 63%, respectively). Surprisingly, δ-nPr-substituted sulfonamido-allenoate could also participate in this transformation, albeit in low yield (**3aj**, 25%), while there is no corresponding product detected for another two δ-alkyl-substituted sulfonamido-allenoates (Cy for **2m** and H for **2n**). Finally, the corresponding hydroisoquinoline derivatives with bulky steric esters (isopropyl for **3ak** and *tert*-butyl for **3al**) were obtained with similar results.

Following the substrate scope evaluation, we attempted to expand the scope of azatrienes. In the presence of P(4-MeC_6H_4)$_3$ and $K_3PO_4·3H_2O$, treatment of 4-bromobenzene sulfonamide **3u** with allenoate **2a** afforded annulation product **3aa** in 73% yield with a dr of 8.3:1, the same ratio as that of the reaction between substrate **1a** and allenoate **2a** (Scheme 4, a). This result demonstrated that the amino group in the product comes from δ-sulfonamido-allenoate **2a**. The reactions between cross-conjugated azatrienes with asymmetric substitution on the two phenyl groups and **2a**, however, did not show good regioselectivity, and they all ended up with mess of isomers, regardless of the difference in the electric effect or steric hindrance (see Scheme S1). Our endeavor to develop an asymmetric edition of this annulation turned out to be in vain, because all tests in the presence of chiral phosphine catalysts resulted in racemization (see Scheme S3).

Then the deuterium labeling experiment was conducted to gain insight into the reaction mechanism (Scheme 4, b). ^{1}H

Scheme 4. Mechanistic Study



NMR analysis showed deuterium incorporation at position C2, and nearly no deuterium was detected at position C1, which may rule out other pathways to the final product (see the Supporting Information for details). Moreover, the result between imine **11** and allenoate **2a** also gave us a clue about how the first annulation reached completion.

On the basis of the results presented above and the related literature precedent,[4a,19] a possible mechanism for this novel reaction is depicted in Figure 2. Intermediate **A**, formed

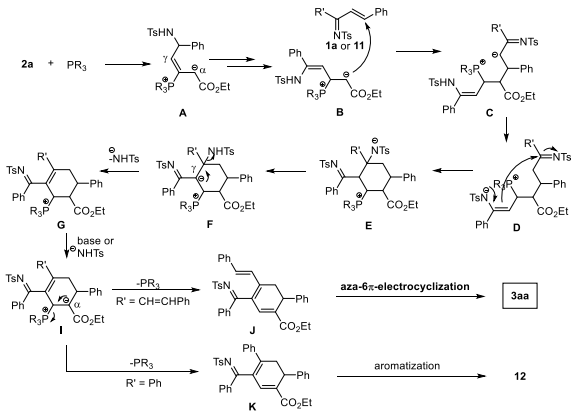


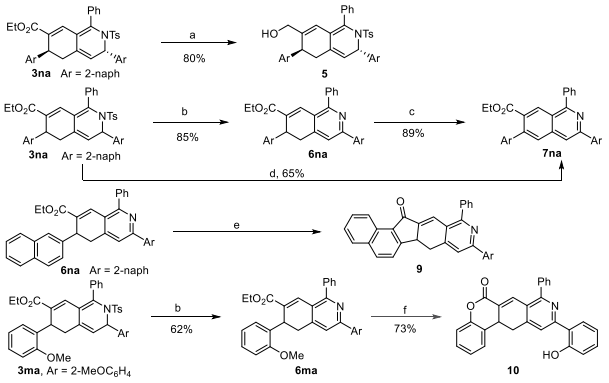
Figure 2. Possible mechanism.

through the nucleophilic attack of the phosphine catalyst on the β-site of δ-sulfonamido-alleno ate **2a**, undergoes proton shifts and isomerization to deliver active species **B** as reported in our previous work.[4a] Next, Michael addition from **B** to **1a** or **11** results in **C**, which undergoes another proton shift to afford **D**. Driven by the eneamine anion, an intramolecular 1,2-addition to the imine takes place, furnishing intermediate **E** through a rarely reported [3+3] process between the cross-conjugated azatriene and allenoate. After subsequent proton shift, the 4-methylbenzenesulfonamide anion serves as both a leaving group and a potential base to abstract the α-site proton, to produce **I**. Then, the release of the PR_3 catalyst affords intermediate **J** (R′ = CH = CHPh) and **K** (R′ = Ph). Finally, **J** undergoes an aza-6π-electrocyclization to deliver **3aa** as the product, while the aromatization of **K** leads to benzannulation product **12**. In our view, the fact that no enantioselectivity priority was observed among all of the trials may reveal the great chance of the presence of an electrocyclization process, due to its inherent quality (see the Supporting Information for asymmetric attempts under chiral phosphine catalysts).

Scheme 5 shows some transformation of tetrahydroisoquinolines **3** to illustrate potential utility. Reduction of the ester of **3na** could be achieved by treatment with lithium aluminum hydride in good yield (80%). In addition, deprotection of the

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Scheme 5. Transformation of Tetrahydroisoquinolines[a]



[a]Reaction conditions: (a) 2 equiv of $LiAlH_4$, dry THF; (b) 6 equiv of SmI_2, dry THF; (c) 1.25 equiv of DDQ, toluene; (d) 3 equiv of DDQ, toluene; (e) (i) 10 equiv of KOH, EtOH, **8**, 84%; (ii) 5 equiv of TfOH, dry CH_2Cl_2, 93%; (f) 2.2 equiv of BBr_3, dry CH_2Cl_2.

nitrogen atom and spontaneous aromatization of **3na** gave rise to dihydroisoquinoline **6na**, which could be easily transformed into isoquinoline derivative **7na** with DDQ. Treatment with 3 equiv of DDQ of product **3na** directly afforded **7na**. We were delighted to find that fused penta- and tetracyclic molecules **9** and **10** could be readily accessible through Friedel−Crafts acylation mediated by TfOH and lactonization mediated by BBr_3, respectively.

In conclusion, we have developed an unprecedented sequential annulation that provides an efficient and facile strategy for the preparation of highly substituted tetrahydroisoquinolines. Reactivities of both substrates, including cross-conjugated azatrienes and δ-sulfonamido-allenoates, were unique and totally different from reported examples. Under mild conditions, two new cycles and three bonds are formed in a one-step process. Furthermore, products could be transformed into dihydro- and isoquinolines and polyfused rings with greatly improved molecular complexity. Mechanistic studies may imply the presence of aza-6π-electrocyclization that has been rarely reported in the annulations mediated by phosphine catalysts.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.0c03725.

Experimental details, characterization data for new compounds, copies of NMR spectra, and the X-ray crystal structure of **3fa** (PDF)

Accession Codes

CCDC 2012531 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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■ DEDICATION

Dedicated to the 100th anniversary of Chemistry at Nankai University.

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